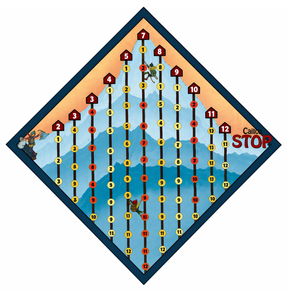
Artificial Intelligence for Can't Stop

From Exact Dynamic Programming to Optimized Heuristics

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1. The Game: Can't Stop

Can't Stop is a push-your-luck dice game for 2–4 players. The board has 11 columns numbered 2–12, each of a different height N_c. Players race to be the first to **complete 3 columns**.

Structure of a turn. A turn consists of a sequence of **rolls**. At each roll:

- Roll **4 dice** and partition them into **2 pairs**. Each pair's sum (ranging from 2 to 12) indicates a column to advance on (either by opening a new column or by continuing in the ones already open).
- At most **3 columns** may be active simultaneously during a turn.
- **Bust**: if no pair sum matches an active or openable column, all temporary progress for the turn is lost.
- **Win**: first player to reach the top of 3 columns.

Col	2	3	4	5	6	7	8	9	10	11	12
N_c	3	5	7	9	11	13	11	9	7	5	3

Table 1: Column heights N_c: outer columns shorter but harder to roll.

Three-phase structure of a roll. At each roll, the player faces three successive sub-problems:

Phase 1 — **Which column(s) to open?** *(if new col. needed)*

↓

Phase 2 — **Stop or Continue?** *(if continuing)*

↓

Phase 3 — **Which dice pairing?** *(if >1 legal option)*

The strategies developed in this project address these sub-problems. For instance, the Rule of 28 (Section 4) handles column opening via a scoring rule and stop/continue via a threshold; the exact DP on 3 columns (Section 2b) gives the optimal stop/continue answer for a fixed column set.

2. Special Cases — Exact Dynamic Programming

Because the full game state space is intractable, we first solve simplified versions exactly, building intuition for the full game.

2a. One Column

Simplified model: single column of height N, each roll succeeds with probability p.

Bellman equation:

$$u_t(s) = \max\Big(\underbrace{s}_{\text{stop}},\ \underbrace{p \cdot u_{t+1}(s+1)}_{\text{continue}}\Big), \quad u_N(s) = s$$

The optimal strategy has a **threshold structure**: stop as soon as $s \geq s^*(p)$.

Closed-form threshold (infinite horizon approximation):

$$s^*(p) = \left\lceil \frac{p}{1-p} \right\rceil$$

Key insight: the higher the success probability, the longer one should push before stopping.

2b. Three Columns

We focus on a reduced board with 3 columns. State: (x, y, z) = temporary heights for each column. The Bellman equation becomes:

$$V_t(x,y,z) = \max\Bigg(\underbrace{\text{stop value}}_{\text{secure progress}},\ \sum_{\omega\in\Omega} p_\omega \Big[\mathbf{1}_{\mathcal{A}=\emptyset} V_{t+1}(x,y,z) + \mathbf{1}_{\mathcal{A}\neq\emptyset} \max_{C\in\mathcal{A}(\omega,x,y,z)} \big(g_t(x,y,z,C) + V_{t+1}(x',y',z')\big)\Big]\Bigg)$$

with:

$$x' = \min(x + \text{mult}(6, C), N_6)$$

(similarly for y', z'), and $|\Omega| = 6^4 = 1296$ equally likely outcomes. $g_t(x,y,z,C)$ is the immediate reward for action C

$\mathcal{A}(\omega, x, y, z)$ is the set of admissible actions in (x, y, z) when we draw w (4 dices)

$V_t(x, y, z)$ is the maximal value when we start the problem from (x, y, z).

General DP formulation:

$$V_t(x) = \max_{u_t\in\mathcal{A}(x)} \mathbb{E}\left[g_t(x,u_t) + V_{t+1}(X_{t+1})\right]$$

Why three reward formulations? The 3-column model is itself an approximation of the full game (columns are fixed, the rest of the board is ignored). We therefore explore different objective functions as approximations of increasing fidelity to the true game, and study how sensitive the resulting policy is to this choice.

Different reward characterizations studied:

1. Reward at each advancement

$$g_t(x,y,z,C) = \text{mult}(6,C) + \text{mult}(7,C) + \text{mult}(8,C)$$

$$\text{stop value} = x + y + z$$

2. Reward only at column completion

$$g_t(x,y,z,C) = 0$$

$$K(x,y,z) = \mathbf{1}_{x=N_6} + \mathbf{1}_{y=N_7} + \mathbf{1}_{z=N_8}$$

3. Minimize expected number of turns

$$V(x,y,z;x',y',z') = \min(\text{stop}, \text{continue})$$

with

$$\text{stop} = 1 + V(x',y',z')$$

$$\text{continue} = \sum_{\omega\in\Omega} p_\omega \Big[\mathbf{1}_{\mathcal{A}=\emptyset}(1 + V(x,y,z)) + \mathbf{1}_{\mathcal{A}\neq\emptyset} \min_{C\in\mathcal{A}} V(x'',y'',z'')\Big]$$

3. Full Game Formulation

For the complete game (with only 1 player), the state becomes:

$$\mathbf{x} = (x_c)_{c\in\{2,\dots,12\}}$$

(secured progress)
and

$$\mathbf{x}' = (x'_c)_{c\in\{2,\dots,12\}}$$

(temporary progress during the current turn).

State-space count

To fully describe a game state, one needs: the secured progress on each column, which (up to 3) columns have been opened this turn, and the temporary height reached on each. The secured state alone yields:

$$\prod_{c=2}^{12}(N_c+1) = 4{\times}6{\times}8{\times}10{\times}12{\times}14{\times}12{\times}10{\times}8{\times}6{\times}4 \approx 7,4\times10^9$$

Adding temporary progress multiplies by a further factor $\sim 10^4$, giving a total of order $\mathbf{6{,}83\times10^{13}}$ states.

Constraint: at most **3 active columns** simultaneously:

$$\left|\mathcal{C}_{\text{act}}(\mathbf{x}') \cup \text{supp}(C)\right| \leq 3$$

General Bellman equation for the full game:

$$\mathcal{V}(\mathbf{x},\mathbf{x}') = \min(\text{stop}, \text{continue})$$

$$\text{stop} = 1 + V(\mathbf{x}')$$

$$\text{continue} = \sum_{\omega\in\Omega} p_\omega \Big[\mathbf{1}_{\mathcal{A}=\emptyset}(1 + V(\mathbf{x})) + \mathbf{1}_{\mathcal{A}\neq\emptyset} \min_{C\in\mathcal{A}(\omega,\mathbf{x},\mathbf{x}')} \mathcal{V}(\mathbf{x},\mathbf{x}''(C))\Big]$$

where:

$$x''_c(C) = \min(x'_c + \text{mult}(c,C), N_c)$$

State-space explosion: $6,83\times10^{13}$
making exact DP intractable for the full game.

Convergence & Computation Times

Problem	No. of states	Conv. time
Max gain (3 col.: 6,7,8)	2.0×10^3	1s
Min turns (3 col.: 6,7,8)	6.6×10^5	16s
Min turns (full game)	6.8×10^{13}	∼30 years

Note: For problems involving the minimization of the average number of iterations, the times shown are for a single step of value iteration. We do 84 iterations, which takes about 20 minutes to converge.

4. Heuristic Strategies

Since exact DP is intractable for the full game, we designed heuristics targeting each of the three turn sub-problems (column selection, stop/continue, dice pairing) — or combinations thereof. All share the interface:

`(progress, temp_progress, legal_actions) ↦ (action, stop?)`

Naive / Baseline Heuristics

Fixed-roll stopping rules

- **Random + Stop@4**: Random legal action, stop after 4 rolls.
- **Score@K** (K = 1 to 5): Score actions favoring central columns + near-completion bonus, stop after exactly K rolls.

Our Heuristics

H1 — Minimize New Openings

Prioritize actions on already active columns. When opening a new column is necessary, choose the one maximizing $\frac{p_c}{\text{remaining}_c}$.

H2 — Probability / Remaining

Action score = $\sum$ mult $\times \frac{p_c}{\text{remaining}_c} - 0.15 \times$ new marker penalty. Balances hit probability and distance to completion.

H3 — Minimize Bust Probability

After each action, compute exact bust probability over all 1296 dice outcomes and choose the safest move.

Literature Heuristics (Glenn & Aloi)

H4 — Rule of 28

Action value: $\sum p_i(6 - |7-i|) - 6 \cdot$ new marker.
Stop when $\sum(s_i+1)(|7-i|+1) +$ bonuses ≥ 28.

H5 — Constant Weights (Genetically Optimized)

Uses pre-optimized column weights (e.g. $w_2 = w_{12} = 7$, $w_7 = 1$) and marker penalty = 6. Stop threshold = 29. Currently the strongest known heuristic.

Our Hybrid Challenger

H6 — DP-3-columns - Min turns (Hybrid)

Column selection via H2 scoring (p_c/remaining$_c$). Stop/continue decision via **exact DP** solved on the 3 currently active columns, with the formulation which minimizes the expected number of turns.

5. Simulation & Results

We built a modular simulator allowing fair comparison of all heuristics on the **same pre-generated dice sequences** (5000 games per heuristic). Reported turn counts include **busts** (a busted turn counts as a played turn).

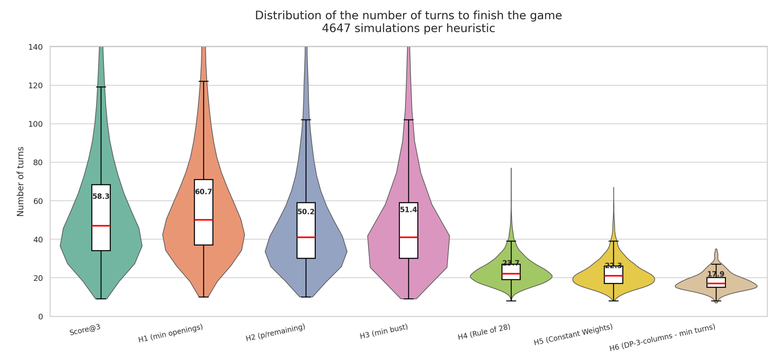


Figure 1: Distribution of turns to finish (5000 simulations per heuristic, busts included). Random+Stop@4 (mean 335.5) excluded for readability. Lower is better.

Heuristic	Mean	Median	P90	Notes
Random + Stop@4	335.5	214.0	763.4	naive baseline
Score@3	58.3	47.0	99.0	
H1 (Min. Openings)	60.7	50.0	100.0	
H2 (p/Remaining)	50.2	41.0	86.0	
H3 (Min. Bust)	51.4	41.0	86.0	
H4 (Rule of 28)	23.7	22.0	33.0	literature
H5 (Const. Weights)	22.3	21.0	32.0	literature
H6 (DP-3-col. min turns)	**17.9**	**17.0**	**24.0**	*best*

Table 2: Performance on 5000 simulations, busts included (lower = better).

6. Conclusions

- Exact DP is **numerically tractable** for 1 and 3 columns; the full game requires heuristics
- A **threshold structure** is provably optimal in the 1-column case; this guides heuristic design
- Our heuristics address the three turn sub-problems in a modular, combinable way
- **H4 and H5** (Rule of 28 and Constant Weights) are faithfully reproduced from the literature, with mean ≈ 23 turns (busts included)
- **H6** (mean: 17.9 turns) outperforms all competitors including both literature benchmarks — a **25% improvement** over H4 — achieved through a purely handcrafted, mathematically grounded heuristic combining probabilistic column scoring with exact DP on the 3 active columns

References

1. Glenn, J., & Aloi, C. (2009). *A Generalized Heuristic for Can't Stop*. Proceedings of the Twenty-Second International FLAIRS Conference.
2. Bellman, R. (1957). *Dynamic Programming*. Princeton University Press.
3. Project implementation in Python 3 (NumPy, Numba). Modular simulator and benchmark framework developed for this project.